



Habib University

EE-426/CE-442 Cryptography and Network Security – Spring 2026

Instructor: Dr. Farhan Khan

Time = 25 minutes

Quiz 01 SOL

Max Points: 20

Instructions:

- i. **Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Attempt the quiz on clean papers with black/blue ink.
- iv. Print your name and HU ID on all sheets.
- v. Turn in your question paper along with your answer sheet(s).
- vi. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	✓
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	

Question 1 [4 points]:

Using the Euclidean algorithm, find the greatest common divisor of the following pair of integers.

- 200 and 51

Based on your result, also comment whether the pair is relatively-prime or not relatively-prime.

Answer:

Step	Quotient	Dividend	Divisor	Remainder
1	3	200	51	47
2	1	51	47	4
3	11	47	4	3
4	1	4	3	1
5	3	3	1	0
6	—	1	0	—

The last non-zero remainder is 1, therefore:

$$\gcd(200, 51) = 1$$

Since the GCD is 1, the numbers are:

Relatively Prime (Coprime)

Question 2 [4 points]:

Using the extended Euclidean algorithm, find the multiplicative inverse of the following integer in $\mathbb{Z}_{26}$. Show all the steps using tabular method.

- 9 (mod 26)

Answer:

Step	Quotient q_i	Dividend	Divisor	Remainder	t_{i-2}	t_{i-1}	t_i
1	2	26	9	8	0	1	-2
2	1	9	8	1	1	-2	3
3	8	8	1	0	-2	3	-26
4	—	1	0	—	3	-26	—

Thus,

$$9^{-1} \equiv 3 \pmod{26}$$

Quick Verification

$$9 \times 3 = 27 \equiv 1 \pmod{26}$$

Question 3 [6 points]:

Check whether it is possible to use the following 3×3 matrix as an encryption key in a HILL CIPHER.

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \pmod{26}$$

Answer:

Hill Cipher Key Matrix (3×3)

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \pmod{26}$$

For a Hill cipher key matrix K to be valid:

$$\gcd(\det(K), 26) = 1$$

Determinant

$$\begin{aligned} \det(K) &= 6(16 \cdot 15 - 10 \cdot 17) - 24(13 \cdot 15 - 10 \cdot 20) + 1(13 \cdot 17 - 16 \cdot 20) \\ &= 6(240 - 170) - 24(195 - 200) + (221 - 320) \\ &= 6(70) - 24(-5) - 99 = 420 + 120 - 99 = 441 \end{aligned}$$

Reduce mod 26:

$$441 \equiv 25 \pmod{26}$$

$$\gcd(25, 26) = 1 \Rightarrow \text{matrix is invertible}$$

Question 4 [6 points]:

Find all solutions to the following set of linear equations:

$$5x + 7y \equiv 3 \pmod{16}$$

$$9x + 4y \equiv 11 \pmod{16}$$

Answer:

Write $AX \equiv B \pmod{16}$:

$$A = \begin{bmatrix} 5 & 7 \\ 9 & 4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 11 \end{bmatrix}$$

Adjoint method for 2×2 :

$$A^{-1} \equiv (\det A)^{-1} \operatorname{adj}(A) \pmod{16}$$

where

$$\det A = ad - bc, \quad \operatorname{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Step 1: Compute $\det(A) \pmod{16}$

$$\det(A) = 5 \cdot 4 - 7 \cdot 9 = 20 - 63 = -43$$

Reduce mod 16:

$$-43 \equiv -43 + 48 = 5 \pmod{16}$$

So

$$\boxed{\det(A) \equiv 5 \pmod{16}}$$

Step 2: Find $(\det A)^{-1} \pmod{16}$

We need $5k \equiv 1 \pmod{16}$.

Try $k = 13$:

$$5 \cdot 13 = 65 \equiv 1 \pmod{16}$$

So:

$$\boxed{5^{-1} \equiv 13 \pmod{16}}$$

Step 3: Compute $\text{adj}(A) \pmod{16}$

$$\text{adj}(A) = \begin{bmatrix} 4 & -7 \\ -9 & 5 \end{bmatrix}$$

Reduce mod 16:

$$-7 \equiv 9, \quad -9 \equiv 7$$

So:

$$\text{adj}(A) \equiv \begin{bmatrix} 4 & 9 \\ 7 & 5 \end{bmatrix} \pmod{16}$$

Step 4: Compute $A^{-1} \equiv 13 \cdot \text{adj}(A) \pmod{16}$

Multiply each entry by 13 and reduce mod 16:

- $13 \cdot 4 = 52 \equiv 4$
- $13 \cdot 9 = 117 \equiv 5$ (since $16 \cdot 7 = 112$, remainder 5)
- $13 \cdot 7 = 91 \equiv 11$ (since $16 \cdot 5 = 80$, remainder 11)
- $13 \cdot 5 = 65 \equiv 1$

So:

$$A^{-1} \equiv \begin{bmatrix} 4 & 5 \\ 11 & 1 \end{bmatrix} \pmod{16}$$

Step 5: Compute $X \equiv A^{-1}B \pmod{16}$

$$\begin{bmatrix} x \\ y \end{bmatrix} \equiv \begin{bmatrix} 4 & 5 \\ 11 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 11 \end{bmatrix} \pmod{16}$$

Compute x

$$x \equiv 4 \cdot 3 + 5 \cdot 11 = 12 + 55 = 67 \equiv 3 \pmod{16}$$

(because $16 \cdot 4 = 64$, remainder 3)

Compute y

$$y \equiv 11 \cdot 3 + 1 \cdot 11 = 33 + 11 = 44 \equiv 12 \pmod{16}$$

(because $16 \cdot 2 = 32$, remainder 12)

Final Answer

$$x \equiv 3 \pmod{16}, \quad y \equiv 12 \pmod{16}$$

Verification by substitution (mod 16)

1. $5x + 7y = 5(3) + 7(12) = 15 + 84 = 99 \equiv 3$
2. $9x + 4y = 9(3) + 4(12) = 27 + 48 = 75 \equiv 11$